

AIBO's First Words – Appendix I

Summary This appendix explains how the paper uses the expectation-maximization (EM) algorithm to discover clusters in unlabeled data. It first reviews the mean and standard deviation of a Gaussian cluster, then shows how to compute the probability that an instance belongs to a given cluster. It next describes the two alternating EM steps: estimating cluster-membership probabilities and re-estimating cluster parameters from those probabilities. The appendix closes by defining likelihood as the optimization target and noting that EM converges to a local maximum, so multiple runs with different initializations may be needed to find the most natural number of clusters.

Appendix I

This appendix briefly describes the EM clustering algorithm used for the experiments in unsupervised learning of classification concepts [1]. The algorithm is based on a statistical model called *finite mixture*. A mixture is a set of k probability distributions, representing k clusters.

In the case of Gaussian distributions, each distribution is determined by two parameters, its mean μ_D and its standard deviation σ_D . If we know that $x_1, x_2, \dots, x_n$ belong to cluster D , then μ_D and σ_D are very easy to compute. For instance, in the simple case in which there is only one numeric attribute x :

$$\mu_D = \frac{\sum_{i=1}^n x_i}{n} \quad (1)$$

$$\sigma_D = \sqrt{\frac{\sum_{i=1}^n (x_i - \mu_D)^2}{n}} \quad (2)$$

If we know μ and σ for the different clusters, it is also easy to compute the probabilities that a given instance comes from each distribution. Given an instance x , the probability that it belongs to cluster D is:

$$P(D | x) = \frac{p(x | \mu_D, \sigma_D)}{\sum_j p(x | \mu_j, \sigma_j)} \quad (3)$$

where $p(x | \mu_D, \sigma_D)$ is the normal distribution function for cluster D :

$$p(x | \mu_D, \sigma_D) = \frac{1}{\sqrt{2\pi}\sigma_D} \exp\left(-\frac{(x - \mu_D)^2}{2\sigma_D^2}\right) \quad (4)$$

The EM algorithm stands for “expectation-maximization.” Given an initial set of distributions, the first step is the calculation of the cluster probabilities (the “expected” class values). The second step is the calculation of the distribution parameters by the “maximization” of the likelihood of the distributions given the data. These two steps are iterated like in a k -means algorithm.

For the estimation of μ_D and σ_D , a slight adjustment must be made compared to equations 1 and 2 due to the fact that only cluster probabilities, not the clusters themselves, are known for each instance. These probabilities act like weights:

$$\mu_D = \frac{\sum_i p_i x_i}{\sum_i p_i} \quad (5)$$

$$\sigma_D = \sqrt{\frac{\sum_i p_i (x_i - \mu_D)^2}{\sum_i p_i}} \quad (6)$$

where the x_i are now all the instances and p_i is the probability that instance i belongs to cluster D .

The overall likelihood of a distribution set is obtained by multiplying the probabilities of the individual instances i :

$$Q = \prod_i P(x_i) \quad (7)$$

where $P(x_i)$ is determined from $p(x | \mu_D, \sigma_D)$. Q is an indicator of the quality of the distribution. Q increases at each iteration. The algorithm stops when the increase of the log-likelihood becomes negligible, for example less than 10^{-6} for ten successive iterations.

The EM algorithm is guaranteed to converge to a maximum but not necessarily to the global maximum. The algorithm could be repeated several times, with a different initial configuration. By varying the number of clusters k , it is possible to determine the one which maximizes the likelihood and thus the “natural” number of clusters.

References

- [1] Ian H. Witten and Eibe Frank. Data Mining. Morgan Kaufmann Publishers, 2000.